

Justification Document: High-Fidelity Upgrades to the pybuildingenergy Pipeline

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Abstract

This document details the scientific reasoning, literature references, and implementation logic for each of the high-fidelity upgrades made to the `pybuildingenergy` pipeline. These modifications address the baseline ISO 52016-1 standard's limitations in dynamic environmental interactions, specifically its tendency to overestimate peak cooling loads.

1 Dynamic Window Properties

- **Scientific Reference:** Zakula et al. (2021) [2] and De Luca et al. (2022) [3].
- **Justification:** The baseline ISO 52016 standard assumes constant values for the thermal transmittance (U_{win}) and the total solar energy transmittance (g_{win}) [2, 3]. Physically, these properties fluctuate significantly based on external wind speed and the solar angle of incidence [2].
- **Implementation Rationale:** To accurately model the transient thermal resistance of the window, the static calculation was replaced with a dynamic time-step formulation [2]:

$$R_{win}(t) = \frac{1}{U_{win}(t)} - R_{si} - R_{se}$$

Additionally, a solar-angle dependent correction factor (F_W) was introduced to prevent the model from over-admitting solar heat at steep sun angles [3]. Using the Karlsson and Roos empirical model, the overall correction factor is computed dynamically as [3]:

$$F_W = \frac{F_{W,diff} I_{sol,diff,t} + F_{W,dir} I_{sol,dir,t} F_{sh,obst,t}}{I_{sol,diff,t} + I_{sol,dir,t} F_{sh,obst,t}}$$

This ensures that the solar transmittance responds physically to the changing boundary conditions at every hourly timestep [2, 3].

2 Escaping Solar Radiation

- **Scientific Reference:** Zakula et al. (2021) [2].
- **Justification:** The standard model assumes that 100% of the solar radiation transmitted through windows is permanently trapped inside the thermal zone [2]. In actual building physics, a portion of this shortwave radiation reflects off internal floors and walls and escapes back out through the glazing [2]. Neglecting this phenomenon artificially inflates the summer cooling load [2].
- **Implementation Rationale:** An internal reflectance factor (ρ_{int}) was applied directly to the internal solar gain accumulator ($\Phi_{sol,zt}$) to mathematically represent escaping shortwave radiation and prevent a perfect heat trap. The adjusted solar heat gain is modeled as:

$$\Phi_{sol,net} = \Phi_{sol,zt} \cdot (1 - \rho_{int})$$

While Zakula et al. (2021) outlines the necessity of modeling this escape mechanism [2], the exact reflectance depends on specific room materials. A standard empirical baseline of $\rho_{int} = 0.15$ (15%) was utilized for the pipeline to approximate this high-fidelity behavior.

3 Split Solar Transmission and Glass Absorption

- **Scientific Reference:** Ballarini et al. (2020) [1].
- **Justification:** The standard applies the total g_{win} directly to the interior air and surface nodes instantaneously [1]. In reality, detailed engines like EnergyPlus split the solar radiation incident on the window into a fraction directly transmitted as light, and a fraction absorbed inside the glass panel [1]. This absorbed heat conducts into the room gradually, creating a necessary thermal lag [1].
- **Implementation Rationale:** The dynamic total solar energy transmittance (g_{dyn}) was mathematically decomposed into a direct transmission coefficient (τ_{win}) and an absorption coefficient (α_{win}):

$$g_{dyn} = \tau_{win} + \alpha_{win}$$

Following the framework highlighted by Ballarini et al. (2020) [1], the absorbed portion (α_{win}) was intentionally routed to the exterior thermal node of the glass, subjecting it to convective wind cooling prior to zone entry. Because the specific split ratios depend on manufacturer glazing properties, the pipeline utilizes a representative approximation for standard double-glazing: $\tau_{win} = 0.85 \cdot g_{dyn}$ and $\alpha_{win} = 0.15 \cdot g_{dyn}$.

4 Variable Air Density for Ventilation

- **Scientific Reference:** Ballarini et al. (2020) [1].
- **Justification:** Ventilation heat transfer calculations in the baseline ISO standard use a constant air density (e.g., 1.204 kg/m^3) [1]. Because hot summer air is less dense, this static assumption overestimates the thermal mass of incoming outdoor air during

peak cooling periods. High-fidelity models evaluate air density dynamically based on air temperature [1].

- **Implementation Rationale:** The Ideal Gas Law was implemented to calculate the exact density of the air mixture dynamically at each time step t . Based on the average absolute temperature of the zone and outdoor air (T_{avg}), the density $\rho_{air}(t)$ is formulated as:

$$\rho_{air}(t) = \frac{P_{atm}}{R_{air} \cdot T_{avg}(t)}$$

Where P_{atm} is standard atmospheric pressure (101325 Pa) and R_{air} is the specific gas constant for air (287.05 J/(kg·K)). This yields a precise volumetric heat capacity for the ventilation flow rate.

5 Non-Linear Long-Wave Sky Radiation

- **Scientific Reference:** De Luca et al. (2022) [3] and Ballarini et al. (2020) [1].
- **Justification:** The baseline model operates on a constant 11 K offset between the outdoor air and the apparent sky temperature [3]. This rigid assumption severely underestimates radiative heat dumping to the sky at night [3]. Detailed models apply a non-linearized formulation using the Stefan-Boltzmann law [1].
- **Implementation Rationale:** The calculation was upgraded to dynamically evaluate the apparent sky temperature (θ_{sky}). The pipeline employs a formulation based on the partial pressure of water vapour ($p_{v,e}$) in Pascals [3]:

$$\theta_{sky} = 18 - 51.6 \cdot e^{-p_{v,e}/1000}$$

Alternatively, utilizing a dry-bulb dependency model (Swinbank), the actual non-linear sky radiation flux (Φ_{sky}) is calculated via the Stefan-Boltzmann law [1]:

$$\Phi_{sky} = F_{sky} \cdot \epsilon \cdot \sigma \cdot (T_{surf}^4 - T_{sky}^4)$$

This ensures that night-time radiant cooling is physically accurate.

6 Transient Heat Accumulation (Thermal Mass Skew)

- **Scientific Reference:** Ballarini et al. (2020) [1].
- **Justification:** The standard’s rigid RC-network node distribution (Class D) distributes thermal mass equally across wall layers [1]. The overall areal heat capacity is fundamentally defined as [1]:

$$\kappa = \sum_{i=1}^{N_{layers}} s_i \rho_i c_i$$

The default distribution fails to accurately mimic real-world thermal lag, causing instantaneous cooling load spikes when solar radiation hits the exterior envelope [1].

- **Implementation Rationale:** To properly model the thermal wave penetration described by Ballarini et al. (2020) [1], a balanced, custom node weight array was assigned to skew the capacitance distribution across the 5 nodes. A typical empirical weighting vector of [0.125, 0.25, 0.25, 0.25, 0.125] was utilized. This provides sufficient thermal mass at the boundary nodes to effectively buffer against sudden exterior solar spikes.

7 Coupling of Thermally Conditioned Zones

- **Scientific Reference:** De Luca et al. (2022) [3].
- **Justification:** The ISO standard evaluates multi-zone buildings without thermal coupling, assuming absolute zero heat transfer between conditioned zones [3]. For residential apartments, heat bleeding through adjacent walls to uncooled corridors or adjacent units is a significant physical mechanism [3].
- **Implementation Rationale:** A dynamic heat bleed calculator was engineered into the internal heat gains loop. It calculates conductive heat transfer ($\Phi_{tr,adj}$) across all shared partitions based on the adjacent zone's temperature (θ_{adj}), the specific transmittance (U), and the shared surface area (A):

$$\Phi_{tr,adj} = \sum U \cdot A \cdot (\theta_{adj} - \theta_{zone})$$

This accurately captures internal heat migration, stabilizing the primary zone's peak loads.

References

- [1] Ballarini, I., Costantino, A., Fabrizio, E., & Corrado, V. (2020). *A Methodology to Investigate the Deviations between Simple and Detailed Dynamic Methods for the Building Energy Performance Assessment*. Energies, 13(23), 6217. <https://doi.org/10.3390/en13236217>
- [2] Zakula, T., Badun, N., Ferdelji, N., & Ugrina, I. (2021). *Framework for the ISO 52016 standard accuracy prediction based on the in-depth sensitivity analysis*. Applied Energy, 298, 117089. <https://doi.org/10.1016/j.apenergy.2021.117089>
- [3] De Luca, G., Bianco Mauthe Degerfeld, F., Ballarini, I., & Corrado, V. (2022). *Improvements of simplified hourly models for the energy assessment of buildings: The application of EN ISO 52016 in Italy*. Energy Reports, 8, 7349-7359. <https://doi.org/10.1016/j.egyr.2022.05.120>